Khachik Sargsyan

Contact Information

Sandia National Laboratories
Combustion Research Facility
P.O. Box 969, MS 9051
Livermore, CA 94551

Mobile: (734) 730-7454
Work: (925) 294-4885
E-mail: ksargsy@sandia.gov
Web: https://www.ksargsyan.net

Computational scientist with a background in applied mathematics and over 20 years of experience in uncertainty quantification, machine learning, statistical analysis, reduced order modeling and numerical methods, applied to complex computational models in a range of disciplines including climate science, statistical physics, materials science, fluid dynamics.

Education

University of Michigan, Ann Arbor, MI, USA

- GPA: 4.05/4.00,
- Thesis: "Mean First Passage Times in the Near-Continuum Limit of Birth-Death Processes",
- Ph.D., Applied and Interdisciplinary Mathematics, August, 2007.

Moscow Institute of Physics and Technology, Moscow, Russian Federation

- GPA: 4.86/5.00,
- B.S., Applied Physics and Mathematics, June, 2001.

Professional Experience

Sandia National Laboratories, Livermore, CA, USA

- *Distinguished Member of Technical Staff* **2024 - present**
  Leading research projects on uncertainty quantification and scientific machine learning applied to computational models of complex physical phenomena.
- *Principal Member of Technical Staff* **2015 - 2024**
- *Senior Member of Technical Staff* **2010 - 2015**
- *Postdoctoral Fellow* **2007 - 2010**

University of Michigan, Ann Arbor, MI, USA

- *Graduate Student Research Assistant* **2003 - 2006**
  Supported by NSF and Michigan Center for Theoretical Physics. Member of the 3-year NSF research project group "Fronts, Fluctuations and Growth".

Moscow Institute of Physics and Technology, Moscow, Russian Federation

- *Research Assistant* **1999 - 2002**
  Institute for Computer Aided Design of RAS and Institute for System Programming of RAS.

Summary

- Over 70 publications in peer-reviewed journals,
- Over 100 research presentations in academic conferences and workshops,
- Estimated ∼500K lines of scientific programming in Python, C/C++, Matlab, Mathematica,
- Key or lead developer in 5 open-source software products,
- Mentoring over 25 graduate students, postdoctoral fellows and early career researchers,
- Teaching and tutoring of a wide range of undergraduate/graduate level math and engineering classes,
- Bronze medal at the International Math Olympiad in 1997,
- Fluent in English, Russian, Armenian. Reading knowledge of French.

Professional Activities

- Lead PI of SNL LDRD-funded 3-year project "Analysis of Neural Networks as Random Dynamical Systems". Developed state-of-the-art methods for improved training and generalization of neural networks. Demonstrated impact on materials science and climate modeling applications.
- Land Modeling uncertainty quantification lead in DOE BER-funded Energy Exascale Earth System Model (E3SM) project, `e3sm.org`. Developed and deployed automated surrogate modeling and calibration tools for a range of E3SM-based studies, including carbon release, crop modeling, vegetation growth, quasi-biennial oscillation.
- Lead of uncertainty quantification and machine learning components in DOE BES-funded Exascale Catalytic Chemistry (ECC) project, `ecc-project.org`. Developed state-of-the-art methods for combining parametric and intrinsic uncertainties for Kinetic Monte-Carlo models. Developed a new machine learning method, minima-preserving neural network (MPNN), to approximate potential energies in catalytic systems.
- Member of DOE FES-funded FusMatML project. Developed and deployed methods and software for augmenting machine-learned potential energy approximations with uncertainty quantification.
- Member of the DOE FASTMath SciDAC institute, focused on applied math algorithms, tools, and software for HPC applications, `fastmath-scidac.llnl.gov`. Developed state-of-the art methods for model error estimation in physical models. The work has been applied across a range of applications.
- Co-PI of DOE ASCR exploratory project "Uncertainty of Physics-Aware Neural Networtks".
- Current collaboration across multitude of projects with *Brown University, Northeastern University, Emory University, University of Michigan, UCLA, NOAA, LANL, ANL, ORNL, LBLL, PNNL.*
- NNSA ASC, 2020-2021: ASC Artificial Machine Intelligence: developed methods for classification of time series for aging detection.
- SNL LDRD, 2017-2019: Subsystem Reduced-Order Modeling and Network Uncertainty Quantification for Rapid, Agile, Extreme-Scale Simulation. Developed network uncertainty quantification strategies for multi-component systems.
- DOD DARPA, 2016-2018: Uncertainty Quantification in LES Computations of Turbulent Multiphase Combustion in a SCRAMJET Engine. Developed and deployed embedded model error estimation techniques.
- DOE BER-ASCR, 2018-2020: Optimization of Sensor Networks for Improving Climate Model Predictions. Project co-PI. Implemented physics-driven machine learning methods for land model surrogate construction and global sensitivity analysis.
- Participated in the development of Sandia Forecast Model for the COVID-19 Epidemic, 2020: SNL forecasts are used in NM Department of Health to assess the epidemic trends and plan for health resource allocation.

Service

- Editorial Board: Journal of Discrete & Continuous Dynamical Systems – S (DCDS-S) and Journal of Machine Learning for Modeling and Computing (JMLMC).
- Organized several sessions at recognized national and international conferences, such as SIAM UQ, SIAM CSE, SIAM AN, AGU, ISBA, USNCCM, with over 100 speakers total. Co-organized Bay Area Scientific Computing Day Workshop, 2018.
- Invited co-chair at the series of workshops on Artificial Intelligence for Earth System Processes, AI4ESP (`ai4esp.org`): Neural Networks and Surrogate Modeling, 2021.
- Invited participation in the US-Norway bilateral AI forum, 2022.
- Co-chair of Foreign National Networking Group (FNNG) at Sandia, 2020-2024.
- Participated or lead over 50 interview panels since 2020. Active participation in WDTS and SULI student intern hiring programs.

- Invited judge for Outstanding Student Participation Award at AGU Fall Meeting 2019.
- Led a SIAM CSE 2023 Affinity Group on Scientific Machine Learning.
- Invited reviewer of multiple proposals to funding agencies: NSF, DOE Office of Science, and Sandia Laboratory Directed Research and Development.
- Invited referee for *Environmental Modelling and Software; Geophysical Research Letters; Scientific Reports; International Journal for Uncertainty Quantification; Computer Methods in Applied Mechanics and Engineering; Structural and Multidisciplinary Optimization; SIAM Journal on Scientific Computing; Journal of Computational Chemistry; International Journal on Chemical Kinetics; SIAM Journal on Uncertainty Quantification; Reliability Engineering and System Safety; Probabilistic Engineering Mechanics; Combustion and Flame; Journal of Computational Science; Mathematics of Climate and Weather Forecasting; Water Resources Research; AIChE Journal; Physics Letters A; Journal of Computational Physics; Journal of Physical Chemistry; Journal of Guidance, Control, and Dynamics; Mathematical Biosciences; AIAA Journal; Multiscale Modeling and Simulation; Physica D; The European Physical Journal B; Computational Geosciences.*
- Member of Society of Industrial and Applied Mathematics (SIAM), American Geophysical Union (AGU), International Society of Bayesian Analysis (ISBA), American Statistical Association (ASA).

Mentorship

- *Postdoctoral (lead advisor):* Nikhil Iyengar (2024-present), Joy Mueller (2022-present), Joshua Hudson (2020-2023), Logan Williams (2022-2023), Varuni Dantanarayana (2020-2021), Vishagan Ratnaswamy (2019-2020), Prashant Rai (2017-2020), Martin Drohmann (2013-2015).
- *Postdoctoral (co-advisor):* Luis Damiano (2023-present), Pieterjan Robbe (2020-2023), Eric Hermes (2018-2022), Xun Huan (2016-2018), Zhen Liu (2012-2014), Thomas Catanach (2017-2018).
- *Sandia Early-Career Mentorship:* Cristian Lacey, Pieterjan Robbe, Moe Khalil, Oscar Diaz-Ibarra, Francesco Rizzi, Tiernan Casey.
- *Graduate Students:* Javier Murgoitio Esandi (2023-2024, U of Southern California), Chase Dwelle (2016-2019, U of Michigan, Ph.D. thesis committee member), Katherine Johnston (2020-2022, U of Washington), Haley Rosso (2021-present, Emory U), Sofia Guzzetti (2018, Emory U).
- *Undergraduate Students:* Jonathan Vo (2018-2019, U of California, Irvine), Sarah Teichman (2016, U of Massachusetts), Joseph Heindel (2016, Seattle Pacific Univ.), Cagan Ozen (2016, Columbia U), Jason Bender (2014-2015, U of Minnesota), Chi Feng (2015, MIT).

Software

- Python Toolkit for Uncertainty Quantification (PyTUQ): a set of tools and workflows for uncertainty quantification, `github.com/sandialabs/pytuq`. Lead developer: developed the library ground up and implemented most of the advanced algorithms.

- Uncertainty Quantification Toolkit (UQTk): a Python/C++ software kit for uncertainty quantification, `sandia.gov/UQToolkit`. Key developer: implemented most of the advanced algorithms present in the library.

- Minima-Preserving Neural Networks (MPNN): a Python library for building neural network approximations to chemistry potentials preserving minima information, `github.com/sandialabs/mpnn`. Sole developer.

- Quantifying Uncertainties in Neural Networks (QUiNN): a Python library centered around various probabilistic wrappers over PyTorch modules in order to provide uncertainty estimation in neural network predictions, `github.com/sandialabs/quinn`. Sole developer.

- FitSNAP: a Python library for building machine learning (ML) potentials based on LAMMPS, `github.com/FitSNAP/FitSNAP`. Developed and implemented uncertainty quantification methods for constructing the ML potentials.

Invited Recent Talks

- "PyTUQ - Python Toolkit for Uncertainty Quantification", Internal Seminar, virtual, September, 2025.
- "Inverse Modeling via Bayesian Inference", Internal Seminar, virtual, April, 2025.
- "Uncertainty Quantification in Computational Science: from Physical Models to Neural Networks", BIRS Workshop Uncertainty Quantification in Neural Network Models, Banff, Canada, February, 2025.
- "Embedded Framework for Model Error Quantification and Propagation", Second USACM Thematic Conference on Uncertainty Quantification for Machine Learning Integrated Physics Modeling (UQ-MLIP 2024), Arlington, VA, August, 2024.
- "Spatio-Temporal Surrogate Construction and Calibration of E3SM Land Model", ESCO 2024 - 9th European Seminar on Computing, (virtual) Plzen, Czechia, June, 2024.
- "Reduced-Dimensional Neural Network Surrogate Construction and Calibration of the E3SM Land Model", CLM PPE Webinar, virtual, February, 2024.
- "Visualizing and Quantifying Uncertainty of Physics-aware Neural Networks", FASTMath All Hands Meeting, Denver, CO, November 2023.
- "Quantifying Uncertainties in Residual Neural Networks and Neural ODEs", UNCECOMP, 5th International Conference on Uncertainty Quantification in Computational Science and Engineering, Athens, Greece, June, 2023.
- "Climate Model Parameterization with Probabilistic Neural Networks", Bilateral AI US-Norway Forum, Oslo, Norway, October, 2022.
- "Training and Generalization of Residual Neural Networks as Discrete Analogues of Neural ODEs", MLDL Workshop, SNL, July, 2022.
- "Model Error Estimation and Uncertainty Quantification of Machine Learning Interatomic Potentials", Error control in first-principles modelling CECAM-EPFL, Lausanne, Switzerland, June, 2022.
- "Bayesian Inference of Interatomic Potentials: Model Errors and Active Learning", MIT CESMIX-UQ, March, 2022.
- "Active Learning and Uncertainty Quantification for Machine Learning Interatomic Potentials", MMLDT/CSET Mechanistic Machine Learning and Digital Twins for Computational Science, Engineering & Technology, (virtual) San Diego, CA, September, 2021.
- "Quantifying and Reducing Uncertainty in the E3SM Land Model Using Surrogate Modeling", E3SM All Hands Monthly Seminar Series, May, 2021.
- "Probabilistic Methods for Forward and Inverse Uncertainty Quantification", Joint BYU/Utah State Department of Mathematics Seminar, October, 2020.
- "Embedded Model Error Methods", Summer School, Department of Mechanical and Aerospace Engineering University of Rome La Sapienza, September, 2020.
- "Overview of Uncertainty Quantification Methods for Complex Models", DOE Climate Modeling PI meeting, November, 2018.
- "Bayesian Framework for Embedded Model Error Representation and Quantification", Joint Statistical Meetings, Vancouver, Canada, August, 2018.
- "Embedded Model Error Representation for Bayesian Model Calibration", Radcliffe Institute for Advanced Study, Harvard University, May, 2018.

- “Probabilistic Methods for Uncertainty Quantification in Computational Models”, MICDE Seminar, University of Michigan, Ann Arbor, April, 2018.

Publications

An up-to-date publication list: `https://scholar.google.com/citations?user=S9st9uEAAAAJ`

[as of 12/25/2025]	All	Since 2020
Citations	3024	1814
h-index	30	23
i10-index	53	38

- L. Damiano, W. Hannah, C. Chen, James J. Benedict, K. Sargsyan, Bert J. Debusschere, Michael S. Eldred, “Improving the Quasi-Biennial Oscillation via a Surrogate-Accelerated Multi-Objective Optimization”, *Journal of Advances in Modeling Earth Systems*, 17:11, p.e2025MS005057, 2025.
- S. Xie, Christopher R. Terai, H. Wang, Q. Tang, J. Fan, S. Burrows, W. Lin, M. Wu, X. Song, Y. Zhang, Mark A. Taylor, J. Golaz, James J. Benedict, C. Chen, Y. Feng, Walter M. Hannah, Z. Ke, Y. Shan, Vincent E. Larson, X. Liu, Michael J. Prather, Jadwiga H. Richter, M. Shrivastava, H. Wan, Guang J. Zhang, K. Zhang, Andrew M. Bradley, P. Cameron-Smith, L. Damiano, Bert J. Debusschere, Aaron S. Donahue, Richard C. Easter, Michael S. Eldred, Brian M. Griffin, O. Guba, Z. Guo, X. Huang, J. Lee, H. Lee, S. Lou, N. Mahfouz, M. Moncrieff, J. Mülmenstädt, Y. Qian, Quazi Z. Rasool, Andrew F. Roberts, S. Santos, K. Sargsyan, J. Shpund, B. Singh, C. Tao, J. Xie, Y. Yang, X. Zeng, C. Zhang, M. Zhang, S. Zhang, T. Zhang, X. Zheng, Robert L. Jacob, L. Leung, Renata B. McCoy, David C. Bader, “The Energy Exascale Earth System Model Version 3: 1. Overview of the Atmospheric Component”, *Journal of Advances in Modeling Earth Systems*, 17:10, p.e2025MS005120, 2025.
- W. Pringle, C. Huang, P. Xue, J. Wang, K. Sargsyan, Miraj B. Kayastha, T. C. Chakraborty, Z. Yang, Y. Qian, Robert D. Hetland, “Coupled Lake-Atmosphere-Land Physics Uncertainties in a Great Lakes Regional Climate Model”, *Journal of Advances in Modeling Earth Systems*, 17:2, p.e2024MS004337, 2025.
- W. Zhou, L. Zhang, A. Sheshukov, J. Wang, M. Zhu, K. Sargsyan, D. Xu, D. Liu, T. Zhang, V. Mazepa, A. Sokolov, V. Valdayskikh, A. Vasiliev, V. Tran, V. Ivanov, “A Novel Framework to Project the Permafrost Fate With Explicit Quantification of Soil Property and Future Climate Uncertainties”, *Journal of Geophysical Research: Earth Surface*, 130:10, p.e2024JF008168, 2025.
- C. Matthews, Michael W. D. Cooper, P. Robbe, T. Casey, K. Sargsyan, Habib N. Najm, G. Pastore, S. Blondel, N. Militello, B. Wirth, A. Levinsky, J. White, T. Gibson, Christopher R. Stanek, David A. Andersson, “Mechanistic Multiscale Uncertainty Propagation in Support of Accelerated Fuel Qualification”, *Nuclear Technology*, p.1–13, 2025.
- C. Galvin, A. Schneider, P. Robbe, M. W. D. Cooper, W. D. Neilson, A. Claisse, C. Matthews, T. A. Casey, K. Sargsyan, H. N. Najm, D. A. Andersson, “Bayesian Calibration of UO2 Creep Rates as a Tool for Accelerated Fuel Qualification”, *Nuclear Technology*, p.1-28, 2025.
- O. Diaz-Ibarra, K. Sargsyan, H. N. Najm, “Surrogate Construction via Weight Parameterization of Residual Neural Networks”, *Computer Methods in Applied Mechanics and Engineering*, 433, p.117468, 2025.
- J. Mueller, K. Sargsyan, C. Daniels, Habib N. Najm, “Polynomial Chaos Surrogate Construction for Random Fields with Parametric Uncertainty”, *SIAM/ASA Journal on Uncertainty Quantification*, 13, 2025.
- Z. Tan, H. Yao, J. Melack, H. Grossart, J. Jansen, S. Balathandayuthabani, K. Sargsyan, L. Leung, “A Lake Biogeochemistry Model for Global Methane Emissions: Model Development, Site-Level Validation, and Global Applicability”, *Journal of Advances in Modeling Earth Systems*, 16:10, p.e2024MS004275, 2024.
- F. Ghahari, K. Sargsyan, E. Taciroglu, “Quantification of Modeling Uncertainty in the Rayleigh Damping Model”, *Earthquake Engineering & Structural Dynamics*, 53:9, p.2950–2956, 2024.

- K. Blöndal, K. Badger, K. Sargsyan, D. H. Bross, B. Ruscic, C. Goldsmith, "Importance Sampling within Configuration Space Integration for Adsorbate Thermophysical Properties: A Case Study for CH3/Ni(111)", *Phys. Chem. Chem. Phys.*, 26, p.17265–17273, 2024.
- L. Williams, K. Sargsyan, A. Rohskopf, Habib N. Najm, "Active learning for SNAP interatomic potentials via Bayesian predictive uncertainty", *Computational Materials Science*, 242, p.113074, 2024.
- F. Ghahari, K. Sargsyan, G. Parker, D. Swensen, M. Celebi, H. Haddadi, E. Taciroglu, "Performance-Based Earthquake Early Warning for Tall Buildings", *Earthquake Spectra*, 40:2, 2024.
- W. Zhou, L. Zhang, A. Sheshukov, J. Wang, M. Zhu, K. Sargsyan, D. Xu, D. Liu, T. Zhang, V. Mazepa, A. Sokolov, V. Valdayskikh, V. Ivanov, "Ground Heat Flux Reconstruction Using Bayesian Uncertainty Quantification Machinery and Surrogate Modeling", *Journal of Geophysical Research - Earth Surface*, 11, e2023EA003435, 2024.
- J. Hudson, M. D'Elia, H. N. Najm, K. Sargsyan, "Measuring Stiffness in Residual Neural Networks", *RAMSES: Reduced order models; Approximation theory; Machine learning; Surrogates, Emulators and Simulators*, p. 153-170, 2024.
- J. Hudson, M. D'Elia, H. Najm, K. Sargsyan, "The Role of Stiffness in Training and Generalization of ResNets", *Journal of Machine Learning for Modeling and Computing*, 4:2, 2023.
- M. Johnson, M. Gierada, E. Hermes, D. Bross, K. Sargsyan, H. Najm, J. Zádor, "Pynta - An automated workflow for calculation of surface and gas-surface kinetics", Accepted, *Journal of Chemical Information and Modeling*, 63:16, 5153–5168, 2023.
- W. Pringle, Z. Burnett, K. Sargsyan, S. Moghimi, E. Myers, "Efficient Probabilistic Prediction and Uncertainty Quantification of Tropical Cyclone–Driven Storm Tides and Inundation", *Artificial Intelligence for the Earth Systems*, 2, e220040, 2023.
- K. Blöndal, K. Sargsyan, David H. Bross, B. Ruscic, C. Goldsmith, "Configuration Space Integration for Adsorbate Partition Functions: The Effect of Anharmonicity on the Thermophysical Properties of CO–Pt(111) and CH3OH–Cu(111)", *ACS Catalysis*, 13:1, pp.19–32, 2023.
- E. Sinha, Katherine V. Calvin, B. Bond-Lamberty, B. Drewniak, D. Ricciuto, K. Sargsyan, Y. Cheng, C. Bernacchi, C. Moore, "Modeling Perennial Bioenergy Crops in the E3SM Land Model (ELMv2)", *Journal of Advances in Modeling Earth Systems*, 15:1, p.e2022MS003171, 2023.
- J. N. Mueller, K. Sargsyan, H. N. Najm, "Polynomial Chaos Surrogate Construction for Stochastic Models with Parametric Uncertainty", *ICASP14: Proceedings of the 14th International Conference on Application of Statistics and Probability in Civil Engineering*, 2023.
- A. Rohskopf, C. Sievers, N. Lubbers, M. A. Cusentino, J. Goff, J. Janssen, M. McCarthy, D. Montes de Oca Zapiain, S. Nikolov, K. Sargsyan, E. Sikorski, L. Williams, D. Sema, A. P. Thompson, M. A. Wood, "FitSNAP: Atomistic machine learning in LAMMPS", *Journal of Open Source Software*, 8(84), 5118, 2023.
- P. Robbe, S. Blondel, T. Casey, A. Lasa, K. Sargsyan, B. Wirth, H. Najm, "Global sensitivity analysis of a coupled multiphysics model to predict surface evolution in fusion plasma–surface interactions", *Computational Materials Science*, 226, 112229, 2023.
- P. Robbe, D. Andersson, L. Bonnet, T. Casey, M. Cooper, C,Matthews, K. Sargsyan, H. Najm, "Bayesian calibration with summary statistics for the prediction of xenon diffusion in UO2 nuclear fuel", *Computational Materials Science*, 225, 112184, 2023.
- E. Hermes, K. Sargsyan, H. N. Najm, J. Zádor, "Sella, an Open-Source Automation-Friendly Molecular Saddle Point Optimizer", *Journal of Chemical Theory and Computation*, 18:11, pp.6974–6988, 2022.
- F. Ghahari, K. Sargsyan, M. Çelebi, E. Taciroglu, "Quantifying modeling uncertainty in simplified beam models for building response prediction", *Structural Control and Health Monitoring*, 29:11, p.e3078, 2022.
- D. Xu, G. Bisht, K. Sargsyan, C. Liao, L. R. Leung, "Using a surrogate-assisted Bayesian framework to calibrate the runoff-generation scheme in the Energy Exascale Earth System Model (E3SM) v1", *Geoscientific Model Development*, 15:12, pp.5021–5043, 2022.

- T.R. Younkin, K. Sargsyan, T. Casey, H.N. Najm, J.M. Canik, D.L. Green, R.P. Doerner, D. Nishijima, M. Baldwin, J. Drobny, D. Curreli, B.D. Wirth, “Quantification of the effect of uncertainty on impurity migration in PISCES-A simulated with GITR”, *Nuclear Fusion*, 62:5, p.056007, 2022.
- K. Blöndal, K. Sargsyan, D. Bross, B. Ruscic, C. Goldsmith, “Adsorbate Partition Functions via Phase Space Integration: Quantifying the Effect of Translational Anharmonicity on Thermodynamic Properties”, *Journal of Physical Chemistry C*, 125:37, pp.20249–20260, 2021.
- V. Y. Ivanov, D. Xu, M. Dwelle, K. Sargsyan, D. B. Wright, N. Katopodes, J. Kim, V. Tran, A. Warnock, S. Fatichi, P. Burlando, E. Caporali, P. Restrepo, B. F. Sanders, M. M. Chaney, A. M. B. Nunes, F. Nardi, E. R. Vivoni, E. Istanbulluoglu, G. Bisht, R. L. Bras, “Breaking Down the Computational Barriers to Real-Time Urban Flood Forecasting”, *Geophysical Research Letters*, 48:20, p.e2021GL093585, 2021.
- B. Kreitz, K. Sargsyan, K. Blöndal, E. J. Mazeau, R. H. West, G. D. Wehinger, T. Turek, C. Goldsmith, “Quantifying the Impact of Parametric Uncertainty on Automatic Mechanism Generation for CO2 Hydrogenation on Ni(111)”, *JACS Au*, 1:10, pp.1656–1673, 2021.
- E. D. Hermes, K. Sargsyan, H. N. Najm, J. Zádor, “Geometry optimization speedup through a geodesic approach to internal coordinates”, *The Journal of Chemical Physics*, 155:9, p.094105, 2021.
- C. Safta, J. Ray, K. Sargsyan, “Characterization of partially observed epidemics through Bayesian inference: application to COVID-19”, *Computational Mechanics*, 66:5, pp.1109–1129, october, 2020.
- V. N. Tran, M. S. Dwelle, K. Sargsyan, V. Ivanov, J. Kim, “A novel modeling framework to secure efficiency and accuracy in real-time ensemble flood forecasting”, *Water Resources Research*, 56:3, p. e2019WR025727, 2020.
- E. Hermes, K. Sargsyan, H. N. Najm, J. Zador, “Accelerated Saddle Point Refinement through Full Exploitation of Partial Hessian Diagonalization”, *Journal of Chemical Theory and Computation*, 15:11, pp.6536–6549, 2019.
- M. C. Dwelle, J. Kim, K. Sargsyan, V. Ivanov, “Streamflow, stomata, and soil pits: sources of inference for complex models with fast, robust uncertainty quantification”, *Advances in Water Resources*, 125, pp.13–31, 2019.
- P. Rai, K. Sargsyan, H. Najm, S. Hirata, “Sparse Low Rank Approximation of Potential Energy Surfaces with Applications in Estimation of AnharmonicZero Point Energies and Frequencies”, *Journal of Mathematical Chemistry*, 57, pp.1732–1754, 2019.
- P. P. Tsilifis, X. Huan, C. Safta, K. Sargsyan, G. Lacaze, J. C. Oefelein, H. N. Najm, R.G. Ghanem, “Compressive sensing adaptation for polynomial chaos expansions”, *Journal of Computational Physics*, 380, pp.29–47, 2019.
- K. Sargsyan, X. Huan, H. N. Najm. “Embedded Model Error Representation for Bayesian Model Calibration”, *International Journal of Uncertainty Quantification*, 9:4, pp. 365–394, 2019.
- D. Ricciuto, K. Sargsyan, P. Thornton, “The Impact of Parametric Uncertainties on Biogeochemistry in the E3SM Land Model”, *Journal of Advances in Modeling Earth Systems*, 10:2, pp.297–319, 2018.
- F. Rizzi, K. Morris, K. Sargsyan, P. Mycek, C. Safta, O. Le Maître, O.M. Knio, B.J. Debusschere, “Exploring the interplay of resilience and energy consumption for a task-based partial differential equations preconditioner”, *Parallel Computing*, 73, p.16–27, 2018.
- O. Cekmer, K. Sargsyan, S. Blondel, H. Najm, D. Bernholdt,, B.D. Wirth, “Uncertainty quantification for incident helium flux in plasma-exposed tungsten”, *International Journal for Uncertainty Quantification*, 8:5, pp.429–446, 2018.
- P. Rai, K. Sargsyan, H. Najm, “Compressed sparse tensor based quadrature for vibrational quantum mechanics integrals”, *Computer Methods in Applied Mechanics and Engineering*, 336, pp.471–484, 2018.
- L. Hakim, G. Lacaze, M. Khalil, K. Sargsyan, H. Najm, J. Oefelein, “Probabilistic parameter estimation in a 2-step chemical kinetics model for n-dodecane jet autoignition”, *Combustion Theory and Modeling*, 22:3, pp.446–466, 2018.
- J. Kenny, K. Sargsyan, S. Knight, G. Michelogiannakis, J. Wilke, “The Pitfalls of Provisioning Exascale Networks: A Trace Replay Analysis for Understanding Communication Performance”,

High Performance Computing, p.269–288, 2018.

- X. Huan, C. Safta, K. Sargsyan, G. Geraci, M. S. Eldred, Z. P. Vane, G. Lacaze, J. C. Oefelein, Habib N. Najm, "Global Sensitivity Analysis and Estimation of Model Error, toward Uncertainty Quantification in Scramjet Computations", *AIAA Journal*, 56:3, pp.1170–1184, 2018.
- X. Huan, C. Safta, K. Sargsyan, Z. P. Vane, G. Lacaze, J. C. Oefelein, H. N. Najm, "Compressive sensing with cross-validation and stop-sampling for sparse polynomial chaos expansions", *SIAM/ASA Journal of Uncertainty Quantification*, 6:2, pp.907–936, 2018.
- F. Rizzi, K. Morris, K. Sargsyan, P. Mycek, C. Safta, O. Le Maitre, O. Knio, B. Debusschere, "Partial differential equations preconditioner resilient to soft and hard faults", *The International Journal of High Performance Computing Applications*, 32:5, pp.658–673, 2018.
- X. Huan, G. Geraci, C. Safta, M.S. Eldred, K. Sargsyan, Z.P. Vane, J.C. Oefelein, H.N. Najm, "Multifidelity Statistical Analysis of Large Eddy Simulations in Scramjet Computations", *AIAA SciTech Forum*, No. AIAA-2018-1180, 2018.
- N. Griffiths, P. Hanson, C. Iversen, A. Malhotra, K. McFarlane, R. Norby, D. Ricciuto, K. Sargsyan, S. Sebestyen, X. Shi, A. Walker, E. Ward, J. Warren, D. Weston, "Temporal and spatial variation in peatland carbon cycling and implications for interpreting responses of an ecosystem-scale warming experiment", *Soil Science Society of America*, 81:6, pp.1668–1688, 2017.
- K. Sargsyan, "Surrogate Models for Uncertainty Propagation and Sensitivity Analysis", "Forward Problems" section, Handbook of Uncertainty Quantification, Springer, 2017.
- B. Debusschere, K. Sargsyan, C. Safta, K. Chowdhary, "The Uncertainty Quantification Toolkit (UQTk)", "Software" section, Handbook of Uncertainty Quantification, 2017.
- P. Mycek, A. Contreras, O. Le Maitre, K. Sargsyan, F. Rizzi, K. Morris, C. Safta, B. Debusschere, O. Knio, "A resilient domain decomposition polynomial chaos solver for uncertain elliptic PDEs", *Computer Physics Communications*, 216, pp.18–34, 2017.
- P. Mycek, F. Rizzi, O. Le Maitre, K. Sargsyan, K. Morris, C. Safta, B. Debusschere, O. Knio, "Discrete A Priori Bounds for the Detection of Corrupted PDE Solutions in Exascale Computations", *SIAM Journal on Scientific Computing*, 39:1, pp.C1–C28, 2017.
- P. Rai, K. Sargsyan, H. Najm, M.R. Hermes, S. Hirata, "Low-rank canonical-tensor decomposition of potential energy surfaces: application to grid-based diagrammatic vibrational Green's function theory", *Molecular Physics*, 115:17-18, pp.2120–2134, 2017.
- M. Khalil, K. Chowdhary, C. Safta, K. Sargsyan, H. N. Najm, "Inference of Reaction Rate Parameters based on Summary Statistics from Experiments", *Proc. Comb. Inst.*, 36:1, pp.699–708, 2017.
- X. Huan, C. Safta, K. Sargsyan, G. Geraci, M. Eldred, Z. Vane, G. Lacaze, J. Oefelein, H. Najm, "Global Sensitivity Analysis and Quantification of Model Error for Large Eddy Simulation in Scramjet Design", *19th AIAA Non-Deterministic Approaches Conference*, No. 2017-1089, 2017.
- K. Morris, F. Rizzi, B. Cook, P. Mycek, O. Le Maitre, O. Knio, K. Sargsyan, K. Dahlgren, B. Debusschere, "Performance scaling variability and energy analysis for a resilient ULFM-based PDE solver", *7th Workshop on Latest Advances in Scalable Algorithms for Large-Scale Systems (ScalA)*, pp.41–48, 2016.
- C. Safta, M. Blaylock, J. Templeton, S. Domino, K. Sargsyan, H. Najm, "Uncertainty Quantification in LES of Channel Flow", *International Journal for Numerical Methods in Fluids*, 83, pp.376–401, 2016.
- K. Morris, F. Rizzi, K. Sargsyan, K. Dahlgren, P. Mycek, C. Safta, O. Le Maitre, O. Knio, B. Debusschere, "Scalability of Partial Differential Equations Preconditioner Resilient to Soft and Hard Faults", *Proceedings of High Performance Computing: 31st International Conference, ISC High Performance 2016*, Frankfurt, Germany, p.469–485, June 19-23, 2016.
- F. Rizzi, K. Morris, K. Sargsyan, P. Mycek, C. Safta, B. Debusschere, O. Le Maitre, O. Knio, "ULFM-MPI implementation of a resilient task-based partial differential equations preconditioner", *Proceedings of the ACM Workshop on Fault-Tolerance for HPC at Extreme Scale*, pp.19–26, 2016.
- J. Ray, Z. Hou, M. Huang, K. Sargsyan, L. Swiler, "Bayesian calibration of the Community Land Model using surrogates", *SIAM/ASA Journal on Uncertainty Quantification*, 3:1, pp.199–233, 2015.

- K. Sargsyan, H. N. Najm, R. Ghanem, "On the Statistical Calibration of Physical Models", *International Journal for Chemical Kinetics*, 47:4, pp. 246–276, 2015.
- F. Rizzi, K. Morris, K. Sargsyan, P. Mycek, C. Safta, O. LeMaitre, O. Knio, B. Debusschere, "Partial differential equations preconditioner resilient to soft and hard faults", *2015 IEEE International Conference on Cluster Computing (CLUSTER)*, pp.552–562, 2015.
- C. Safta, D. Ricciuto, K. Sargsyan, B. Debusschere, H.N. Najm, M. Williams, P. Thornton, "Global Sensitivity Analysis, Probabilistic Calibration, and Predictive Assessment for the Data Assimilation Linked Ecosystem Carbon Model", *Geosci. Model Dev.*, 8, pp. 1899–1918, 2015.
- K. Sargsyan, F. Rizzi, P. Mycek, C. Safta, K. Morris, H. N. Najm, O. Le Maître, O. Knio, B. Debusschere, "Fault Resilient Domain Decomposition Preconditioner for PDEs", *SIAM Journal on Scientific Computing*, 37:5, pp. 2317–2345, 2015.
- J.D. Jakeman, M.S. Eldred, K. Sargsyan, "Enhancing l1-minimization estimates of polynomial chaos expansions using basis selection", *Journal of Computational Physics*, 289, pp. 18–34, 2015.
- C. Safta, K. Sargsyan, B. Debusschere, H. N. Najm, "Hybrid discrete/continuum algorithms for stochastic reaction networks", *Journal of Computational Physics*, 281:0, pp.177–198, 2015.
- C. Safta, K. Sargsyan, H.N. Najm, K. Chowdhary, B. Debusschere, L.P. Swiler, M.S. Eldred, "Probabilistic Methods for Sensitivity Analysis and Calibration of Computer Models in the NASA Challenge Problem", *Journal of Aerospace Information Systems*, 12, pp.219–234, 2015.
- M. Salloum, K. Sargsyan, R. Jones, H. N. Najm, B. Debusschere, "Quantifying Sampling Noise and Parametric Uncertainty in Atomistic-to-Continuum Simulations Using Surrogate Models", *Multiscale Modeling & Simulation*, 13:3, pp.953–976, 2015.
- H. N. Najm, R.D. Berry, C. Safta, K. Sargsyan, B.J. Debusschere, "Data Free Inference of Uncertain Parameters in Chemical Models", *International Journal for Uncertainty Quantification*, 4:2, pp.111–132, 2014.
- J. Wilke, J. Kenny, M. Drohmann, K. Sargsyan, "Towards a Universal Toolkit for Quantifying Simulation Error via both Bayesian Inference and Model Reduction Strategies", *Workshop on Modeling and Simulation of Systems and Applications (MODSIM)*, September, 2014.
- K. Sargsyan, C. Safta, H. N. Najm, B. Debusschere, D. Ricciuto, P. Thornton, "Dimensionality Reduction for Complex Models via Bayesian Compressive Sensing", *International Journal of Uncertainty Quantification*, 4:1, pp.63–93, 2014.
- C. Safta, K. Chowdhary, K. Sargsyan, H.N. Najm, B. Debusschere, L.P. Swiler, M.S. Eldred, "Uncertainty Quantification Methods for Model Calibration, Validation, and Risk Analysis", *AIAA-2014-1497, 16-th AIAA Non-Deterministic Approaches Conference*, January, 2014.
- J. Prager, H.N. Najm, K. Sargsyan, C. Safta, W.J. Pitz, "Uncertainty Quantification of Reaction Mechanisms Accounting for Correlations Introduced by Rate Rules and Fitted Arrhenius Parameters", *Combustion and Flame*, 160, pp.1583–1593, 2013.
- J. Wilke, K. Sargsyan, J. Kenny, B. Debusschere, H. Najm, G. Hendry, "Validation and uncertainty assessment of extreme-scale HPC simulation through Bayesian inference", *European Conference on Parallel Processing*, pp.41–52, August, 2013.
- K. Sargsyan, C. Safta, B. Debusschere, H. Najm, "Multiparameter Spectral Representation of Noise-Induced Competence in Bacillus Subtilis", *IEEE/ACM Trans. Comp. Biol. and Bioinf.*, 9:6, pp. 1709–1723, 2012.
- K. Sargsyan, C. Safta, B. Debusschere and H. N. Najm, "Uncertainty Quantification given Discontinuous Model Response and a Limited Number of Model Runs". *SIAM Journal on Scientific Computing* 34:1, pp. 44–64, 2012.
- F. Rizzi, O.M. Knio, H.N. Najm, B.J. Debusschere, K. Sargsyan, M. Salloum, H. Adalsteinsson, "Uncertainty Quantification in MD Simulations. Part I: Stochastic reformulation of the forward problem", *SIAM J. Multiscale Model. Simul*, 10:4, pp.1428–1459, 2012.
- F. Rizzi, O.M. Knio, H.N. Najm, B.J. Debusschere, K. Sargsyan, M. Salloum, H. Adalsteinsson, "Uncertainty Quantification in MD Simulations. Part II: Inference of force-field parameters", *SIAM J. Multiscale Model. Simul*, 10:4, pp.1460–1492, 2012.

- M. Salloum, K. Sargsyan, R. Jones, B.J. Debusschere, H.N. Najm, H. Adalsteinsson, "A Stochastic Multiscale Coupling Scheme to Account for Sampling Noise in Atomistic-to-Continuum Simulations", *Multiscale Modeling & Simulation*, 10:2, pp.550–584, 2012.
- M. Kalita, K. Sargsyan, B. Tian, A. Paulucci-Holthauzen, H. Najm, B. Debusschere, A. Brasier, "Sources of cell-to-cell variability in canonical nuclear factor-κ B (NF-κ B) signaling pathway inferred from single cell dynamic images", *Journal of Biological Chemistry*, 286:43, pp.37741–37757, 2011.
- K. Sargsyan, B. Debusschere, H. N. Najm and O. Le Maître, "Spectral Representation and Reduced Order Modeling of the Dynamics of Stochastic Reaction Networks via Adaptive Data Partitioning". *SIAM Journal on Scientific Computing*, 31, pp.4395-4421, 2010.
- K. Sargsyan, B. Debusschere, H. N. Najm and Y. Marzouk, "Bayesian Inference of Spectral Expansions for Predictability Assessment in Stochastic Reaction Networks". *Journal of Computational and Theoretical Nanoscience*, 6:10, 2009.
- K. Sargsyan, C. Safta, B. Debusschere, H. Najm, "Uncertainty Quantification in the Presence of Limited Climate Model Data with Discontinuities", *ICDMW '09: Proceedings of the 2009 IEEE International Conference on Data Mining Workshops*, pp.241–247, December, 2009.
- C. Doering, K. Sargsyan, L. M. Sander, E. Vanden-Eijnden, "Asymptotics of rare events in birth-death processes bypassing the exact solutions". *Journal of Physics: Condensed Matter*, 19:6, p.5145, 2007.
- C. Doering, K. Sargsyan and L. Sander, "Extinction times for birth-death processes: exact results, continuum asymptotics, and the failure of the Fokker-Planck approximation". *SIAM Journal on Multiscale Modeling and Simulation*, 3:2, pp.283-299, 2006.
- C. Doering, K. Sargsyan and P. Smereka, "A numerical method for some stochastic differential equations with multiplicative noise". *Physics Letters A*, 344:2-4, pp.149-155, 2005.